

Interactive Assembly Installation

Proposal review: critique, additions, references, and technical feasibility

Summary

The core concept — a tabletop installation where the user physically assembles a small architectural object three different ways (masonry, 3D-printed concrete, modular prefab) with a projector overlaying live LCA data on each piece — is strong, with real white space in the field. This document collects six critique points, eight ranked additions, ten reference projects, and a blunt feasibility verdict on the proposed Rhino / Grasshopper / Anemone stack.

1. Critical analysis — six weak points to address

These are the points a tutor or critic will press on. Better to preempt them than be caught.

1. Three full builds is a UX trap.

Around fifteen minutes reassembling the same wall three times and visitors will fatigue or skip rounds. Better: physically build ONE method end-to-end, play the other two as projected animations on adjacent zones. Same comparison, fraction of the attention cost.

2. The proposal flattens what isn't flat.

A 3D-printed concrete wall and a brick wall are not “the same wall” — they differ structurally, thermally, acoustically, aesthetically. Reducing them to CO₂ / hours / cost is exactly the LCA tunnel vision the project should be critiquing, not reproducing.

3. The numbers are contested.

Environmental Product Declarations (EPDs) for cement and brick vary massively by source. Cite one figure per method and the whole installation rests on a number any expert can dismantle in thirty seconds. Fix: project ranges, with sources visible on the piece.

4. “You feel it because you built it” is the project's own marketing — handle with care.

The visitor feels the time it took them to follow projector arrows, not the time a real mason would take. The labor-hours mapping is symbolic, not embodied. Either own that explicitly (it's a metaphor, not a measurement) or it collapses under serious critique.

5. The three methods are suspiciously canonical.

Masonry / 3DP / prefab is the lecture-slide trio. Swap one for **reclaimed / reused brick** and the whole story flips — reused materials have near-zero embodied carbon and would make 3DP look bad. Sharper and more provocative than the safe choice currently proposed.

6. No politics.

The proposal is silent on who LOSES jobs in each method — 3DP eliminates skilled mason work, prefab moves jobs from local sites to factories in other countries. For an architecture school the political economy of construction is the hot button. Leaving it out is a missed opportunity.

2. Additions to propose — ranked by impact-per-effort

Top picks (worth fighting for in the meeting)

1. Material-origin map projected ON each piece.

When the user places a brick, a tiny world map projects directly onto the brick's surface: clay from X, fired in Y, shipped via Z. For 3DP: cement source, sand source, printer origin. For prefab: factory and truck route. This is the strongest “data becomes physical” moment in the whole installation — supply chains made visible at the scale of a hand.

2. Context dial.

A physical knob that changes the climate zone or grid carbon (Beirut vs Zurich vs Lagos). Same pieces, comparison flips. Forces visitors to conclude: no method is universally best. This kills the lazy “which one wins?” question and replaces it with “wins for whom, where?” — far more honest.

3. Carbon-budget clock ticking on the table.

A continuously projected counter showing the construction industry's remaining 1.5°C carbon budget. Each piece placed ticks it down by its share. Cheap to implement, makes stakes visceral without text.

4. Reused-brick fourth method.

Salvaged brick — embodied carbon near zero. It will outperform 3DP on every metric. Forces visitors to ask why we don't already do this, opening the conversation to policy and supply chains rather than tech fetishism. **This is the single biggest narrative upgrade.**

5. Failure states.

Misplace a brick → projector animates a small collapse. Misalign a 3DP layer → whole layer wasted (cost ticks up). Prefab tolerance off → leaking gap. Lets visitors feel the failure cost of each method, not just the success cost.

Optional / nice to have

- **Disassembly mode.** Build it once, then “deconstruct” and project recyclability or landfill fate of each piece. The circular economy story.

- **Soundscape per method.** Trowel scraping vs pump motor vs factory clang and truck reversing. Zero extra hardware, large atmospheric payoff.

- **End-of-experience vote.** A physical button. System logs choices and projects a running tally — “47 visitors picked masonry, 12 picked 3DP.” Social dimension for free.

3. Reference projects

All real, all relevant. Pull the closest three or four into the proposal deck.

Guided assembly — direct precedents

- bullet **Augmented Bricklaying** — Gramazio Kohler Research, ETH Zurich (2018–20). AR-guided masons placing bricks for complex facades. The direct ancestor of this idea. gramaziokohler.arch.ethz.ch
- bullet **Steampunk Pavilion** — Fologram + Gwyllim Jahn, Tallinn Architecture Biennale 2019. Untrained builders assembling a curved timber pavilion entirely from AR instructions. Public installation that proved guided assembly works with the public. fologram.com/projects/steampunk-pavilion
- bullet **Fologram** — Grasshopper-native AR toolkit. Worth knowing about even if not adopted; it is the commercial product of this whole research line. fologram.com
- bullet **RoMA (Robotic Modeling Assistant)** — Huaishu Peng, Cornell / Autodesk, CHI 2018. Robot and AR guide a user assembling a physical model in real time. The academic citation for “overlaid-on-piece guidance.”

Tangible / physical data visualization

- bullet **inFORM** — Hiroshi Ishii / MIT Tangible Media Group, 2013. The canonical “data you can touch” reference.
- bullet **Domestic Data Streamers** — Barcelona. Participatory physical dataviz where visitors aggregate into sculptural data. Directly relevant to “make LCA felt.” domesticstreamers.com

LCA / embodied carbon

- bullet **Material Cultures** — London, 2021—. Bio-based comparison demonstrators. Same comparison logic as this proposal, but full-scale and not interactive. materialcultures.org
- bullet **EPFL Structural Xploration Lab** — Corentin Fivet. Pavilions built from reclaimed components with embodied-carbon as design driver. Cite this if you take the reused-brick suggestion. epfl.ch/labs/sxl
- bullet **Striatus Bridge** — Zaha Hadid Computation + Block Research Group + ETH, 2021. The “what 3D-printed concrete actually looks like” image for the slides.
- bullet **TECLA** — WASP + Mario Cucinella, 2021. Earth-based 3DP housing, explicitly framed around embodied carbon.

Key gap (and an opportunity)

There is no strong precedent for a tabletop **interactive** LCA comparison installation. Guided assembly exists. Tangible dataviz exists. LCA comparison demonstrators exist. Nobody has done all three together at table scale. That is white space — this concept has territory to claim, and the proposal can say so explicitly.

4. Technical feasibility — blunt verdict

Rhino + Grasshopper + Anemone + webcam + projector is a **fragile** stack for a public installation. Concretely:

- Grasshopper is single-threaded; Anemone stalls the canvas while looping — bad for real-time response.
- Webcam piece-tracking inside Grasshopper (via Firefly or Quokka) is slow and very light-sensitive — and the projector's own light contaminates the camera feed, which is a known nightmare.
- Projection-mapping calibration onto small physical pieces drifts constantly. Grasshopper gives zero help here.
- Students on this stack typically burn the final week fighting canvas freezes instead of finishing the experience.

Recommended alternative

TouchDesigner. Native projection mapping, computer vision via OpenCV TOPs, gallery-stable for long runs. Use Rhino / Grasshopper only to design geometry and run the LCA math offline; export to TouchDesigner for the runtime, with an OSC bridge if the installation needs to talk back to Grasshopper. Second choice: Unity + MadMapper.

If the group insists on staying in Grasshopper

- Use **Firefly** (not Anemone) for the camera input.
- Put **fiducial markers** (ArUco) on every physical piece so detection does not depend on color.
- Run the projector from a separate machine to avoid blocking the canvas.

5. Draft message for the group chat

Paste-ready. Six points, ordered from concept to stack so the conversation flows naturally.

Hey — went deep on this, here is what I would push for:

- 1. Strong concept but the three-builds loop will exhaust visitors. Suggest physically building only ONE method, with the other two playing as projected animations on adjacent zones. Same comparison, way better attention budget.*
- 2. Add a **reused-brick fourth method**. Embodied carbon near zero, it will outperform 3DP on every metric, which turns the whole installation into a sharper political question instead of a tech showcase.*
- 3. Two cheap additions with big payoff: a **material-origin map** projected directly on each piece (clay from X, cement from Y) and a **carbon-budget clock** ticking on the table as we build. Both make data felt without extra hardware.*
- 4. LCA numbers should be projected as **ranges with sources**, not single figures — EPDs vary wildly and a single number kills our credibility under questioning.*
- 5. On the stack: I would push back on Anemone as the runtime. Grasshopper is single-threaded and Anemone stalls the canvas; combined with webcam tracking under projector light it is a known fragile combo. **TouchDesigner** is the standard for this kind of installation — does projection mapping and CV natively, much more stable. Keep Rhino and Grasshopper for geometry and LCA math offline. Worth at least discussing before we commit.*
- 6. References for our slides: Gramazio Kohler's Augmented Bricklaying (ETH), Fologram's Steampunk Pavilion (Tallinn 2019), and EPFL Fivet's reuse demonstrators. Worth noting in the proposal: there is no precedent for an interactive tabletop LCA comparison installation — that is our white space.*

Free Wed or Thu to meet.